

Map the traps

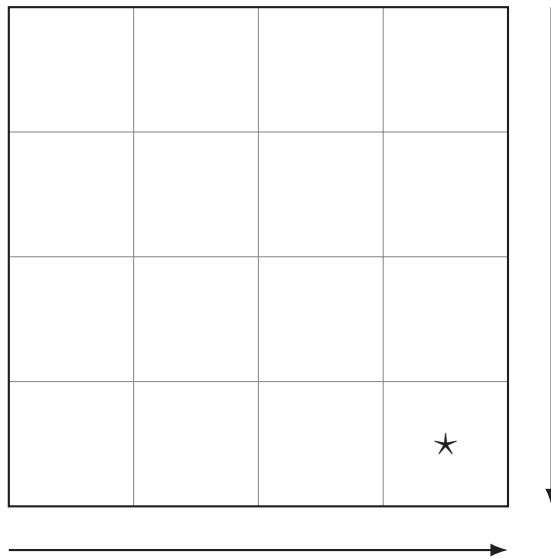
Name: _____

Date: _____

Share one token. Move right OR down any positive number of squares. No diagonal moves. Reach the star to win.

1. Play, then make a map.

Try different starts. Mark a square T if the player who receives it must lose against perfect replies. Mark W if that player can force a win. Use ? while unsure.



Treat the star as already finished: the next player has no move. Start your map near it. A single lost game does not prove T!

2. A claim that survives every reply.

Choose one T. Show that **every** move from it gives the next player a W. Then choose one W and show **one** move to a T.

My two squares and reasons:

Try it with objects. Explain by pointing, talking, or drawing.

The board is two piles

Name: _____

Date: _____

3. Count what is left.

Put the token 3 squares left and 2 squares above the star. Make a pile of 3 and a pile of 2. A right move shrinks the first pile; a down move shrinks the second.

Pile rule: remove any positive number from ONE pile. Take the last counter to win. Play both versions and match their moves.

4. Prove a rule for every rectangle.

Which two-pile positions are traps? Explain these two things with counters:

1. Every move from a trap breaks your pattern.
2. From any other position, you can make the pattern.

My pattern and why both statements hold:

5. Add a third pile.

Keep the same pile rule. Try $(1, 1, 1)$, then $(1, 2, 3)$. Does “two matching piles” still guarantee a trap? List all first moves from $(1, 2, 3)$ and find a winning reply to each.

A surprise or a new conjecture:

Make a puzzle: Choose a two-pile start with exactly one winning move. Let a partner test it.

Try it with objects. Explain by pointing, talking, or drawing.